

NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Extension 2

Assessment Task 2

Term 2: 2009

Name: _____

Mathematics Class: 12MZ _____

Time Allowed: 60 minutes + 2 minutes reading time

Available Marks: 46

Instructions:

- Start each question on a new page.
- Put your name on the top of each page.
- Attempt all four questions.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Write on one side of the page only. Do not work in columns.
- Each question will be collected separately.
- If you do not attempt a question, submit a blank page with your name and the question number clearly displayed.

Question	1	2	3	4	Total
E3		/12	b/6		/18
E4	/11		a/5		/16
E9				/12	/12
	/11	/12	/11	/12	/46
					%

Question 1. (11 marks)**Marks**

- (a) It is given that $3+i$ is a zero of $P(x) = x^3 + rx + 60$ where r is real.
- (i) State why $3-i$ is also a zero of $P(x)$. 1
- (ii) Factorise $P(x)$ over the real numbers. 2
- (iii) Find the value of r . 1
- (b) Decompose $\frac{3x^2 - 7x + 4}{(x-3)(x^2+1)}$ into partial fractions over the real field. 3
- (c) The equation $x^3 - 3x - 1 = 0$ has roots α, β, γ .
- (i) Show that none of α, β, γ are integers. 2
- (ii) Form the polynomial equation with roots $\alpha^2, \beta^2, \gamma^2$. 2

Question 2. (12 marks)

- (a) $P(x, y)$, $Q(8, 0)$ and $R(-8, 0)$ are three points in the number plane such that $PQ + PR = 24$ units.
- (i) Explain why P lies on an ellipse. 1
- (ii) Find the equation of the locus of P in standard form. 3
- (b) (i) The hyperbola $x^2 - y^2 = 8$ is rotated 45° anticlockwise until it coincides with $xy = c^2$. Write down the value of c . (No proof is required.) 1
- (ii) Draw a neat sketch showing this information using the above value of c , showing clearly the coordinates of the foci and the equations of the directrices of $xy = c^2$. 3
- (c) The point $P(a \cos \theta, b \sin \theta)$ lies on one end of the latus rectum drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
(The latus rectum is the focal chord perpendicular to the major axis)
- (i) Show that $\cos \theta = e$ where e is the eccentricity. 1
- (ii) Show that the length of the latus rectum is given by $\frac{2b^2}{a}$. 3

Question 3. (11 marks)**Marks**

- (a) (i) $P(x) = 0$ is a polynomial with a root of multiplicity m . Show that $P'(x) = 0$ has a root of multiplicity $(m-1)$. 2
- (ii) The polynomial $\alpha x^{n+1} + \beta x^n + 1$ is divisible by $(x-1)^2$.
Show that $\alpha = n$ and $\beta = -(1+n)$. 3
- (b) $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ are points on the hyperbola $xy = 4$. M is the midpoint of the chord PQ .
 P and Q move on the hyperbola so that the chord PQ always passes through the point $R(4,2)$.
- (i) Show that the equation of the chord PQ is given by $x + pqy = 2(p+q)$ 2
- (ii) Show that $pq = p + q - 2$ 1
- (iii) Show that the locus of M is given by $y = 1 + \frac{2}{x-2}$ 3

Question 4. (12 marks)

- (a) $P(x_1, y_1)$ is a variable point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
- (i) Show that the equation of the normal at P is given by $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$ 3
- (ii) The tangent and normal at P meet the y -axis at G and H respectively and S is a focus of the hyperbola. Show that $\angle GSH$ is a right angle.
(You may assume the equation of the tangent at P is given by $\frac{x_1x}{a^2} - \frac{y_1y}{b^2} = 1$) 3
- (b) (i) Show the solutions to the equation $z^9 - 1 = 0$ on the Argand plane. 1
- (ii) Show that the solutions to $z^6 + z^3 + 1 = 0$ are contained in the solutions to $z^9 - 1 = 0$. 1
- (iii) Hence factorise $z^6 + z^3 + 1 = 0$ into real quadratic factors. 2
- (iv) Show that $(1 - \cos \frac{2\pi}{9})(1 - \cos \frac{4\pi}{9})(1 + \cos \frac{\pi}{9}) = \frac{3}{8}$. 2

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End of paper

EXTENSION 2: 2009 TASK 2 Solutions

Question 1

(a) (i) coefficients real
 $\therefore$ roots in conjugate pairs
 $\therefore (3-i)$ a zero

(ii) $P(x) = x^3 + 1x + 60$
 $= [x^2 - (3-i+3+i)x + (3-i)(3+i)]$
 $\times (x-b)$
 $= (x^2 - 6x + 10)(x-b)$
 $\therefore -10b = 60$

$$\therefore b = -6$$

$$\therefore P(x) = (x^2 - 6x + 10)(x + 6)$$

(iii) $x = -6$ a zero
 $\therefore P(-6) = -6^3 - 6r + 60 = 0$
 $6r = -6^3 + 60$
 $r = -6^2 + 10$
 $r = -26$

(b) $\frac{3x^2 - 7x + 4}{(x-3)(x^2+1)} = \frac{a}{x-3} + \frac{bx+c}{x^2+1}$

$$3x^2 - 7x + 4 = a(x^2+1) + (bx+c)(x-3)$$

$x=3$ $27 - 21 + 4 = 10a$

$$a = 1$$

coeff x : $3 = a + b$

$$\therefore b = 2$$

$x=0$ $4 = a - 3c$

$$3c = 1 - 4$$

$$c = -1$$

$$\therefore \frac{3x^2 - 7x + 4}{(x-3)(x^2+1)} = \frac{1}{x-3} + \frac{2x-1}{x^2+1}$$

(c) $x^3 - 3x - 1 = 0$

(i) monic, integer coefficients
 $\therefore$ any integer zeros divide constant term (-1)

ie only possible integer zeros $x = \pm 1$

$$P(1) = 1 - 3 - 1 = -3 \neq 0$$

$$P(-1) = -1 - 3 - 1 = -5 \neq 0$$

$\therefore$ No integer zeros.

(ii) x^2, x^2, x^2 satisfy $P(\sqrt{x}) = 0$

$$\sqrt{x}^3 - 3\sqrt{x} - 1 = 0$$

$$\sqrt{x}(x-3) = 1$$

$$x(x-3)^2 = 1$$

$$x(x^2 - 6x + 9) = 1$$

$$x^3 - 6x^2 + 9x - 1 = 0$$

Question 2

(a) $PQ + PR = 24$

(i) ellipse since sum of distances to 2 fixed points is a constant

(ii) $2a = 24$

$$a = 12$$

$$ae = 8 \text{ since } F(\pm 8, 0)$$

$$e = \frac{2}{3}$$

$$\therefore b^2 = a^2(1 - e^2)$$

$$b^2 = 144(1 - \frac{4}{9})$$

$$b^2 = 144 \times \frac{5}{9}$$

$$= 80$$

$\therefore$ Equation $\frac{x^2}{144} + \frac{y^2}{80} = 1$

Exat. 2: 2009 Task 2 Solution

①

Question 2 (c) (i) $x^2 - y^2 = 8$

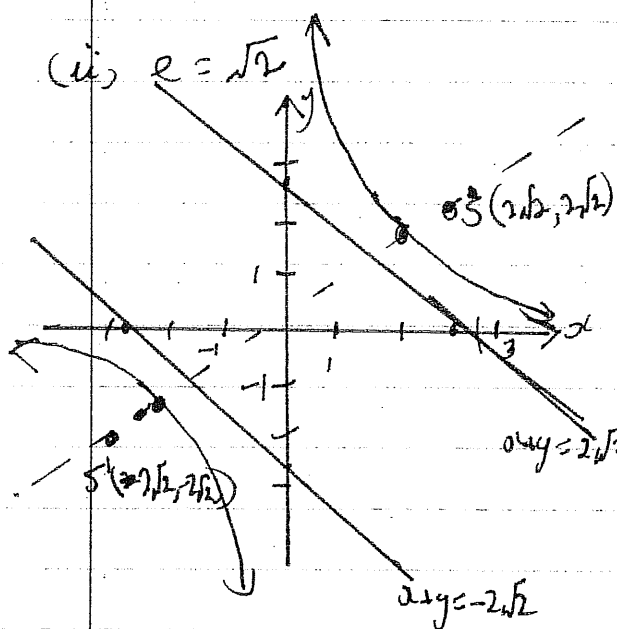
$$xy = c^2$$

$$\therefore c^2 + c^2 = \sqrt{8^2}$$

$$\therefore 2c^2 = 8$$

$$c^2 = 4$$

$$\therefore xy = 4$$



(c) (i) $a \cos \theta = ae$

$$\therefore \cos \theta = e$$

$$\begin{aligned} \text{(ii) length } L.R. &= 2b \sin \theta \\ &= 2b \sqrt{1 - \cos^2 \theta} \\ &= 2b \sqrt{1 - e^2} \end{aligned}$$

$$\text{but } b^2 = a^2(1 - e^2)$$

$$\therefore 1 - e^2 = \frac{b^2}{a^2}$$

$$\therefore \text{length} = 2b \cdot \frac{b}{a}$$

$$= \frac{2b^2}{a}$$

Question 3

(a) (i) let $p(x) = (x-\alpha)^m Q(x)$

$$p'(x) = m(x-\alpha)^{m-1} Q(x) + Q'(x)(x-\alpha)^m$$

$$= (x-\alpha)^{m-1} (mQ(x) + Q'(x)(x-\alpha))$$

$$= (x-\alpha)^{m-1} \times \text{polynomial}$$

$\therefore p'(x)$ has root multiplicity $(m-1)$

(ii) $P(x) = \alpha x^{n+1} + \beta x^n + 1$

$(x-1)^2$ factor $\therefore p(1) = p'(1) = 0$

$$p(1) = \alpha + \beta + 1 = 0 \quad \text{--- ①}$$

$$p'(x) = \alpha(n+1)x^n + \beta nx^{n-1}$$

$$p'(1) = (n+1)\alpha + \beta n = 0 \quad \text{--- ②}$$

$$\text{①} \times n: \quad \alpha n + \beta n + n = 0 \quad \text{--- ③}$$

$$\text{③} - \text{②}: \quad \alpha n - \alpha n - \alpha + n = 0$$

$$\therefore \alpha = n$$

$$\text{sub into ①} \quad n + \beta + 1 = 0$$

$$\beta = -(1+n)$$

$$(b) m_{pq} = \left(\frac{2}{q} - \frac{2}{p} \right) \div (2q - 2p)$$

$$= \frac{p-q}{pq} \times \frac{1}{2-p}$$

$$= -\frac{1}{pq}$$

Equation

$$y - \frac{2}{p} = -\frac{1}{pq} (x - 2p)$$

$$pqy - 2q = -x + 2p$$

$$x + pqy = 2p + 2q$$

$$x + pqy = 2(p+q)$$

Ex 2 : 2009 : Task 2 solution

Question 3 (b) through (4, 2)

$$\therefore 4 + 2pq = 2(p+q)$$

$$2 + pq = p+q$$

$$pq = p+q-2$$

$$(iii) M \left[\frac{2(p+q)}{2}, \frac{2(\frac{1}{p} + \frac{1}{q})}{2} \right]$$

$$x = p+q$$

$$y = \frac{1}{p} + \frac{1}{q}$$

$$g = \frac{q+p}{pq}$$

$$= \frac{q+p}{p+q-2}$$

$$\text{but } x = p+q$$

$$y = \frac{0x}{x-2}$$

$$= \frac{0x-2+2}{x-2}$$

$$= 1 + \frac{2}{x-2}$$

$$\text{at } G \text{ } y=0 \quad \frac{ax}{a^2} - \frac{y^2}{b^2} = 1$$

$$y = -\frac{b^2}{y_1}$$

$$S(ae, 0) \quad G(0, \frac{-b^2}{y_1}) \quad H(0, \frac{y_1(a^2+b^2)}{b^2})$$

$$m_{SG} = \frac{b^2}{y_1} / ae = \frac{b^2}{ae y_1}$$

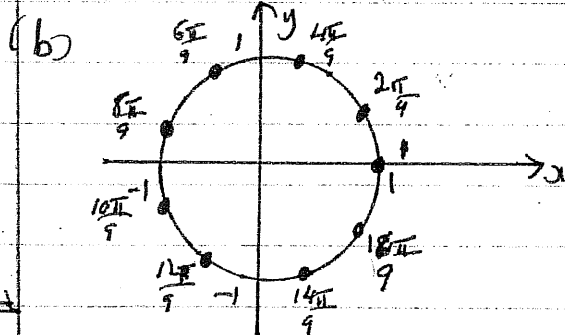
$$m_{SH} = \frac{y_1(a^2+b^2)}{b^2} / -ae = \frac{y_1(a^2+b^2)}{-b^2 ae}$$

$$\text{but } a^2+b^2 = a^2 e^2$$

$$m_{SH} = \frac{y_1(a^2 e^2)}{-b^2 ae} = -\frac{ae y_1}{b^2}$$

$$\therefore m_{SG} \times m_{SH} = -1 \therefore SG \perp SH$$

$$\therefore \angle GSH = 90^\circ$$



$$(ii) z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1)$$

$$\therefore z^6 + z^3 + 1 \text{ contained in solutions to } z^9 = 1$$

$$\text{which do not satisfy } z^3 = 1 \text{ i.e. } \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$(iii) z^6 + z^3 + 1 = 0$$

$$\text{roots with args. } \pm \frac{2\pi}{9}, \pm \frac{4\pi}{9}, \pm \frac{8\pi}{9}$$

$$\therefore z^6 + z^3 + 1 = [z^2 - 2\cos \frac{2\pi}{9} z + 1][z^2 - 2\cos \frac{4\pi}{9} z + 1][z^2 - 2\cos \frac{8\pi}{9} z + 1]$$

$$(iv) |z| = 1$$

$$1+1+1 = (2-2\cos \frac{2\pi}{9})(2-2\cos \frac{4\pi}{9})(2-2\cos \frac{8\pi}{9})$$

$$\therefore (1-\cos \frac{2\pi}{9})(1-\cos \frac{4\pi}{9})(1-\cos \frac{8\pi}{9}) = \frac{3}{8}$$

$$\text{but } \cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}$$

$$\therefore (1-\cos \frac{2\pi}{9})(1-\cos \frac{4\pi}{9})(1+\cos \frac{\pi}{9}) = \frac{3}{8}$$

Question 4

$$(a) \frac{ax}{a^2} - \frac{y^2}{b^2} = 1$$

$$(i) \frac{ax}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dx}{dy} = \frac{b^2 x}{a^2 y}$$

$$m_N = \frac{-a^2 y_1}{b^2 x_1}$$

$$\text{equation: } y - y_1 = \frac{-a^2 y_1}{b^2 x_1} (x - x_1)$$

$$b^2 a_1 y - b^2 x_1 y_1 = -a^2 x_1 y_1 + a^2 x_1 y$$

$$a^2 x_1 y_1 + b^2 y_1 x_1 = a^2 x_1 y_1 (a^2 + b^2)$$

$$\frac{a^2 ax}{a_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$(ii) \text{ at } H: x=0$$

$$\therefore y = \frac{y_1(a^2+b^2)}{b^2}$$